

Gilberto G. Briscoe-Martinez

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SKILLS

MACHINE LEARNING

PyTorch, diffusion policies, imitation learning, ACT, VLA and VLM fine-tuning (LoRA), graph encoders (GAT, HGT), evaluation and ablation design, CUDA, NumPy

ROBOTICS

Motion planning, trajectory optimization, compliant and contact-rich control, non-prehensile manipulation, multi-robot coordination, ROS 2, Isaac Sim, OpenCV

SOFTWARE

Python, C++, CMake

PLATFORMS

Franka Panda, UR5e, UR20, NVIDIA Jetson and Orin, industrial induction cells

EDUCATION

UNIVERSITY OF COLORADO BOULDER

PH.D. IN COMPUTER SCIENCE
Expected May 2027 | Boulder, CO
HIRO Group
Advisor: Prof. A. Roncone

UNIVERSITY OF TEXAS AT AUSTIN

B.S. IN AEROSPACE ENGINEERING
May 2021 | Austin, TX
Human Centered Robotics Lab
(2018–2021)

AWARDS

2022 – 2026 NASA NSTGRO Fellowship
(Grant 80NSSC22K1211)

EXPERIENCE

HIRO GROUP, CU BOULDER | PHD RESEARCHER & NASA NSTGRO FELLOW

Aug 2021 – Present | Boulder, CO

- Built **DEFT**, a diffusion policy that recovers from any joint failure on a 7-DoF Franka arm without retraining: 99.5% task success vs 42.4% for RRT, 10/10 real-world tasks with multiple joints locked, trained on 4.7M trajectories (ICRA 2026)
- Designed a fail-active controller for a two-arm package induction cell that reroutes work around faulty arms and conveyors instead of halting: a diffusion policy conditioned on a graph of each component's remaining capacity; $10\times$ the throughput of the same policy without the graph under severe faults (ICRA 2027, under review)
- Tested whether the language model in a vision-language-action (VLA) policy affects the robot's actions, using logs from a production sorting arm; perturbing its output barely changed the picks, so the policy was mostly behavior cloning
- Developed pushing and poking primitives that complete tasks after an end-effector failure: reachable workspace up to 79% larger, 88.9% task success (IROS 2024)
- Co-developed **CERG**, a controller that bounds the arm's energy during contact (IFAC WC 2026, oral); mentor junior PhD students and undergraduates

PLUS ONE ROBOTICS | ROBOTICS AI/ML RESEARCHER (PART-TIME)

June 2025 – Present | Boulder, CO

- Built the proof of concept for learned motion planning on the package induction cell, aimed at cutting new-site spin-up from 8–12 weeks to 6–10 by removing most post-install tuning; the CTO reassigned about 6 engineers to productionize it
- Fine-tuned a VLA policy on 20 hours of test-cell data; after 3 months it runs the cell at 65% of the classical system's throughput and retries failed picks without the hand-coded confirmation step the classical system needs
- Designed the policy architecture the team builds on: a scalable multi-model task and motion planning approach, physically-grounded through a graph representation of the cell, so one policy runs across multiple cells with different arms and conveyors
- Built a half-size test cell with robots and conveyors in a closed loop, and a five-conveyor, two-arm Isaac Sim model that has generated 150+ hours of training data; test on the full-scale cell in San Antonio every two months

HUMAN CENTERED ROBOTICS LAB, UT AUSTIN | UNDERGRADUATE RESEARCHER

Jan 2018 – May 2021 | UT Austin Villa RoboCup@Home | Austin, TX

- Trained object-recognition models for the team's home-service robot (2019) and implemented data augmentation techniques that improved recognition performance from 60% to >95% in real world; co-authored the team's RoboCup 2019 report
- Built the robot's navigation, task planning, and behavior-tree task execution with Prof. Luis Sentis and Prof. Justin Hart; co-authored the team's AAAI AI-HRI 2019 report

METECs, NASA JOHNSON SPACE CENTER | ROBOTICS SOFTWARE INTERN

May 2020 – Aug 2020 | Dexterous Robotics Lab, Houston, TX

- Wrote affordance templates, reusable task models of grasp points and motions, so Robonaut 2, NASA's International Space Station humanoid, could open latches and doors

JOHNS HOPKINS APL | ROBOTICS SOFTWARE ENGINEER (INTERN)

June 2021 – July 2021 | Remote, Laurel, MD

- Built a ROS 1 and ROS 2 simulator with Gazebo models of real buildings so the team could keep developing between tests on the physical robots

PUBLICATIONS

- **ICRA 2027** (under review) — *What Can My Neighbor Still Do? Fail-Active Control for Process-Coupled Robot Cells via Graph-Conditioned Diffusion*
- **ICRA 2027** (under review) — *PBD-R: Rigid-Body Dynamics in Particle-Based Simulation*
- **ICRA 2026** — *DEFT: diffusion policy for joint-failure recovery*
- **IFAC WC 2026** (oral) — *CERG: Compliant Explicit Reference Governor*
- **IROS 2024** — *task recovery after end-effector failure via pushing and poking*

Issued U.S. Patent **10,339,827**. Full list on Google Scholar.